土力学与基础工程

Soil Mechanics and Foundation Engineering

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教师介绍 Introduction about myself

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Research interests: **frost heave; salt expansion**

Q&A

Ask me questions before& after school

Resume

He has been the principal investigator of 3 National Natural Science Foundation projects, and published over 110 academic papers and obtained more than 50 national patents.

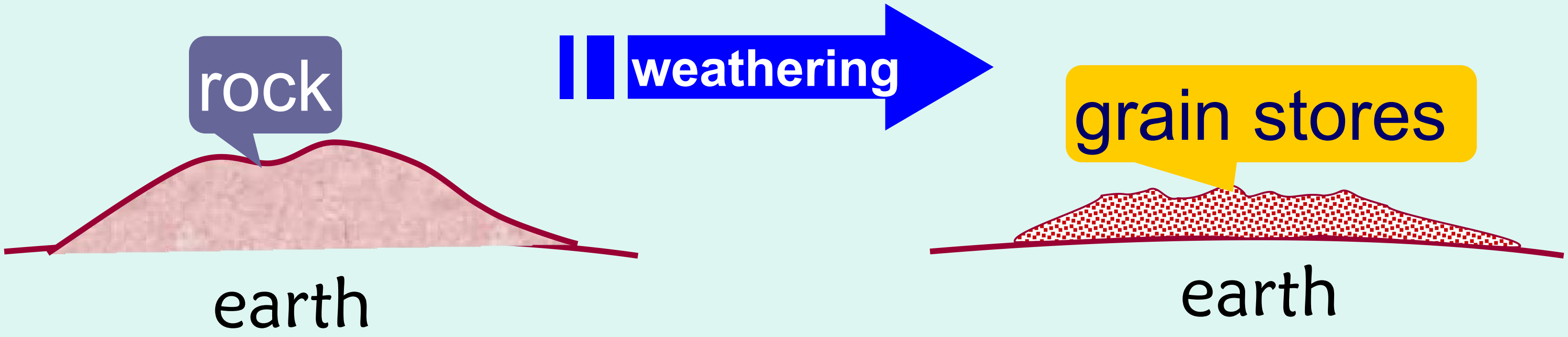
- Professor, doctoral supervisor
- Graduated from the University of Chinese Academy of Sciences
- Editorial Board of ASCE Journal of Cold Regions Engineering
- Candidate for academic and technical leaders in Sichuan Province
- Sichuan Civil Engineering and Architecture Youth Science and Technology Award
- The First Prize of Science and Technology Progress of Gansu Province

INDUCTION

- What’s the soil?
- Characteristics of soil ?
- Characteristics of soil mechanics?
- Why to learn soil mechanics ?
- What’s included by soil mechanics?
- What’s is foundation?
- How to learn soil mechanics well?

Introduction

✵What's the soil ?



soil:

The particle accumulations covered on the ground surface which is clastic、uncemented or weak cementation ， formated from the whole rock of earth surface which subjected to long-term weathering in air.

Introduction

✵ **Characteristics of soil ？**

Introduction

✵ Characteristics of soil ?

Weathering product of rock

↓

particulate characteristic

↓

Discrete medium ⇒

- easily to be deformed after being forced
- volume changes is hole changes chiefly
- shear-cut deformation is caused by relative displacement
- low strength

Introduction

✵Characteristics of soil ?

- solid phase—soil framework
- liquid phase—water
- vapor phase—gas

tri-phase system

discrete medium

- undertake through both soil framework and hole medium after being forced
- existing complex interacting
- hole fluid flow

Introduction

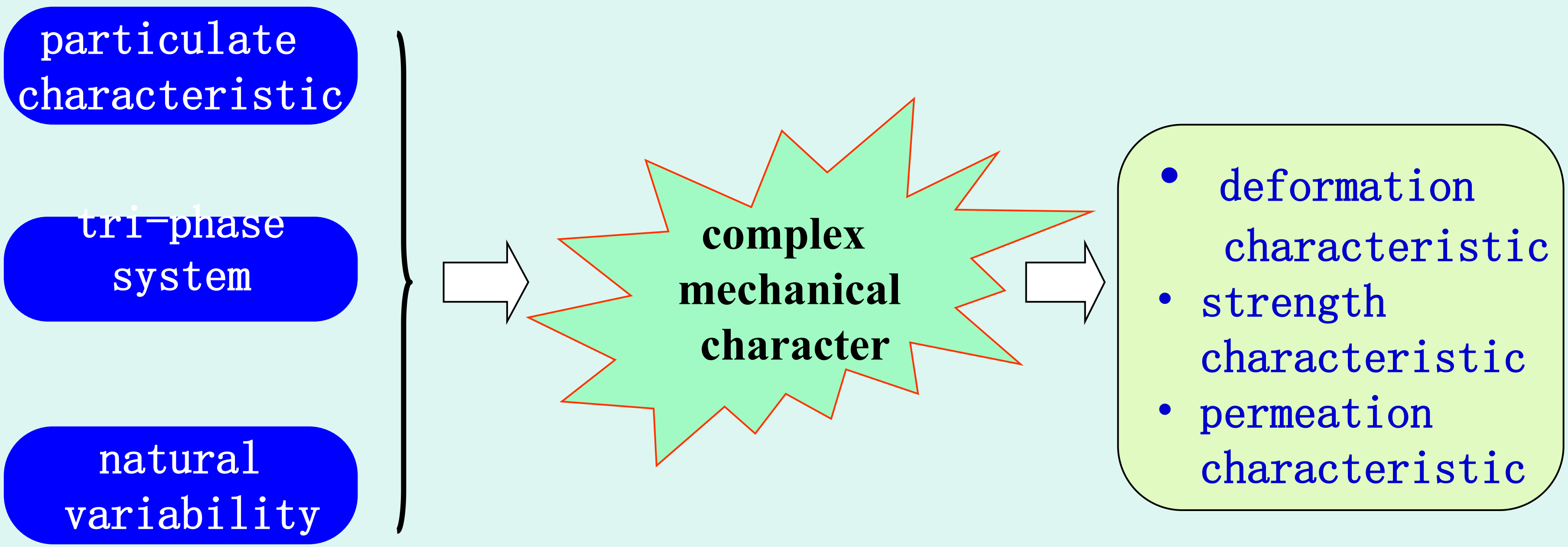
✵ Characteristics of soil ?

natural product

natural variability

- **heterogeneity**
- **aeolotropism**
- **designability**
- **space-time variability**

Introduction ✵ Characteristics of soil ?



Introduction

❈ Characteristics of soil ?

<table>
<tr><th>subject</th><th colspan="2">subject investigated</th></tr>
<tr><td>theory mechanics</td><td colspan="2">particle or rigid body</td></tr>
<tr><td>material mechanics</td><td>single elastic member bar
（pole、shaft、beam）</td><td rowspan="3">continuous solid body</td></tr>
<tr><td>structural mechanics</td><td>member bar structure
some elastic member bar</td></tr>
<tr><td>elastic mechanics</td><td>elastic body structure
or slab-shell structure</td></tr>
<tr><td>hydraulics</td><td>incompressible continuous fluid body （water）</td><td>continuous fluid body</td></tr>
<tr><td>soil mechanics</td><td>natural tri-phase particulate debris</td><td>particulate material</td></tr>
</table>

Introduction

Characteristics of soil ?

elementary knowledge of continuous medium mechanics

describe the theory of particulate

soil mechanics

deformation, strength and permeability of soil and their engineering problems

Introduction Characteristics of soil ?

soil mechanics phylogeny

soil mechanics become a significant sign of independent subject

Terzaghi is a founder of soil mechanics

1776 Coulomb intensity law, soil pressure theory

1856 Darcy law

1857 Rankine new soil pressure theory

1925 Terzaghi effective stress theory and permeate-consolidate theory

1936 international soil mechanics and foundation engineering conference of the first time

1949 the rise of soil mechanics study in China

Introduction

❈Why to learn soil mechanics ?

1. soil has widespread project application

2. many engineering problems about soil

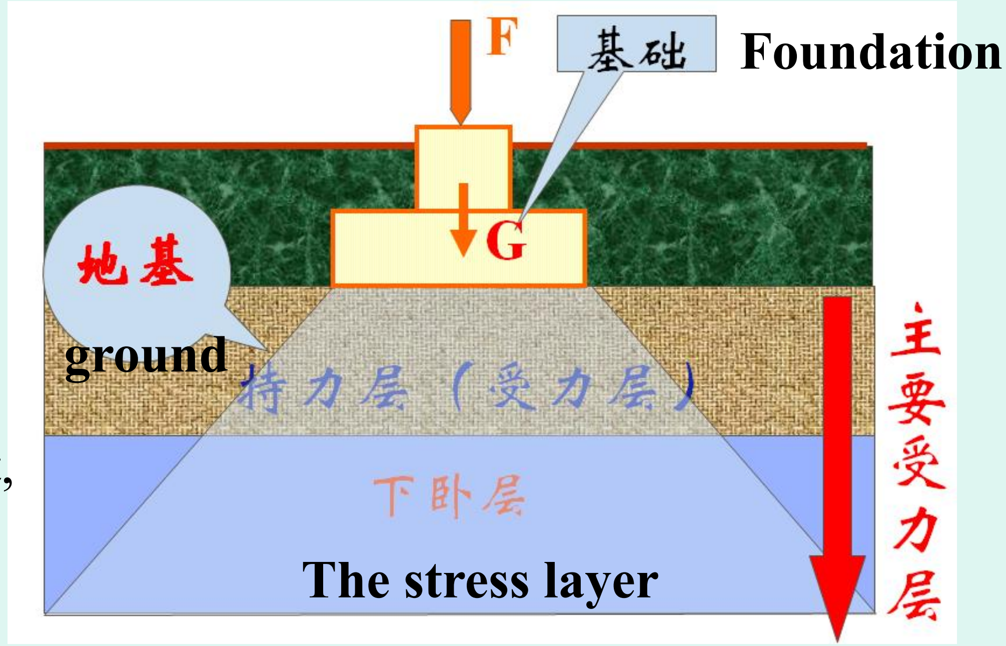
Introduction Why to learn soil mechanics？

1． project application of soil

- building materials of civil engineering
- groundwork building foundation
- structure environment

Introduction

Concept of soil mechanics



Foundation: structural component, transfer the various effects of the superstructure to ground.

Ground: The soil or rock mass that supports the foundation.

An applied science that using the general **principles of mechanics** to study the physics, chemistry and mechanical property of soil; **engineering properties of soil** under the action of such factors as load, water, temperature, et al..

Introduction　Why to learn soil mechanics ?

2. engineering problems about soil

nonuniform deformation

Influence on traffic stability

特征：青藏公路在路基反复冻融作用下，产生不均匀膨胀与沉降，严重影响行车稳定性，并每年发生多起交通事故。

铺设块碎石 pave crushed-rock

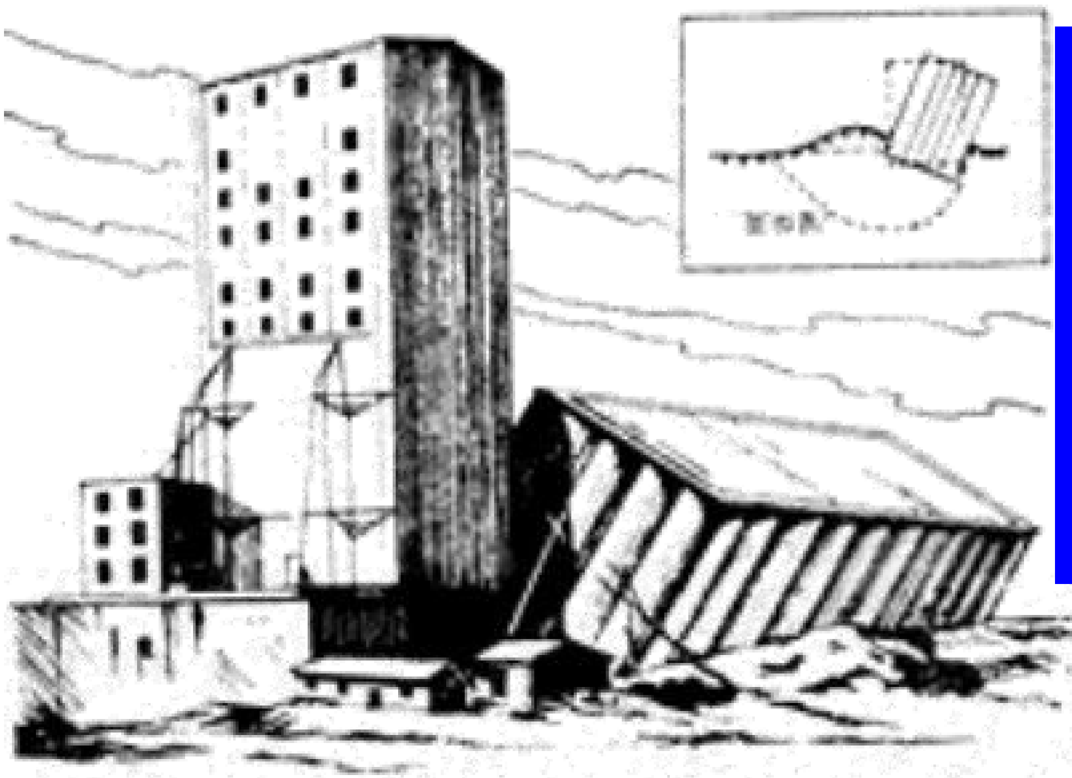
通风管 ventilating pipe

热棒 heat-bar

铺设保温板 pave insulation board

Tlangsikang barn of Canada

Survey: length:59.4m, width:23.5m, high:31.0m, total:65 cylinder barns. steel concrete raft flank foundation, thick:61cm, deep:3.66m. start in 1911, end in 1913, self-weight:20000T.



accident:

lade grain in Sep. 1911,31822 T grain was laded in Oct. 1911,

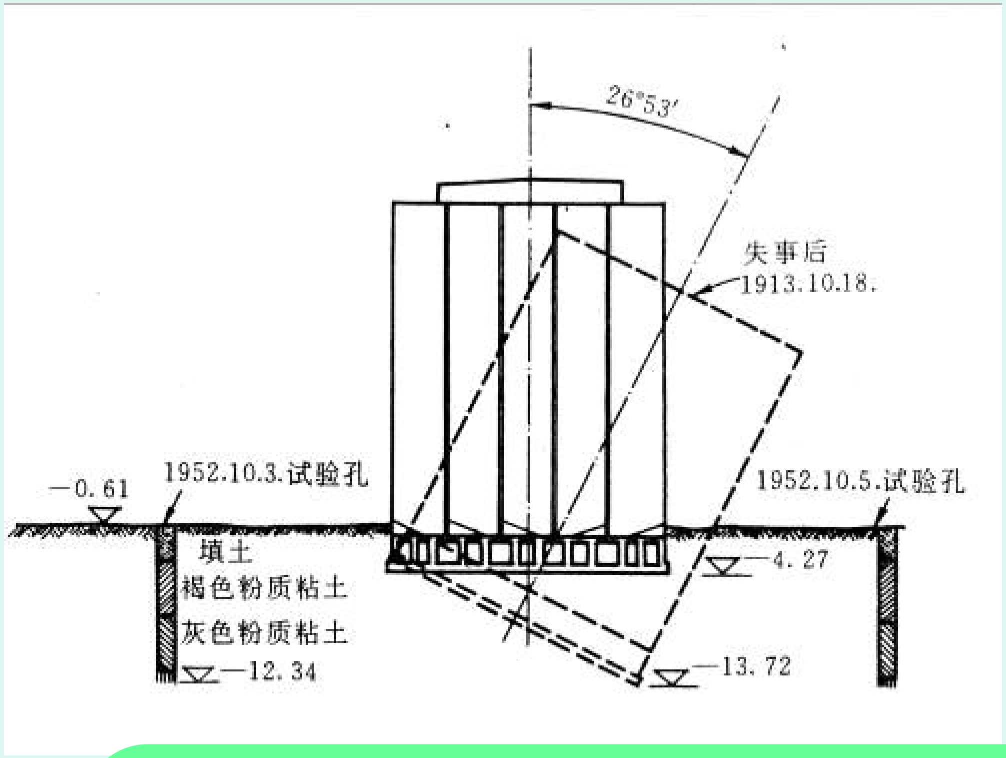
- 1 hour verticals ettlement:30.5cm
- 24 hours stilt:26° 53′
- western subsidence:7.32m
  eastern arise:1.52m
- upside steel concrete silo
  without damage

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Tlangsikang barn in Canada



reason:
Groundwork soil wasn' t investigated beforehand, according to the soiling-testing result of close structure foundation ditch, using the way of calculating the ground bearing capacity into this barn. By way of the reconnaissance testing and calculating in 1952, the practical ground bearing capacity is far less than the foundation pressure when the barn is destroyed. Thus, the intension damage will be took place because of overloading, which leads the barn-foundation to glide.

transacting:
After the accident, made more than 70 concrete piers supporting on bedrocks under the barn, and made barn body corrected gradually using 388 50t jacks and support-system, but the high reduced by 4 m.

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Hongkong Baocheng slip

one morning in july 1972, near the Baocheng load in HongKong, twenty thousant cubic miters native soil slide on the slope , so big silde mass just striked a tall building called Baocheng which colided with a five-floor building on the way of it's slide. so small buidling was instantly damaged and died 67 persons.

reason:

Intension of native soil is lower on the hill side, additional, the rainwater influent make its intension lower and lower much, which leads the sliding force of soil body exceed the intension of soil, thus the soil body slides on the hillside.

Hongkong build city in 1900,establish soil project office in 1977
Island 1972 Po Shan landslip (~ 20,000 m³)(67 die、20 injure)


Early 1972 before landslip


June 1972 after landslip

Engineering problems about soil

foundation liquefaction of Bansheng large earthquake

Shenghu dock:
Lateral deformation and settlement happened after the earthquake caused large-area sandy soil foundation liquefaction. A mass of building had been broken down or suffered severe damage.

liquefaction: a phenomenon which the blending foundation loses its intension and becomes in flowage because of vibrating load

Introduction

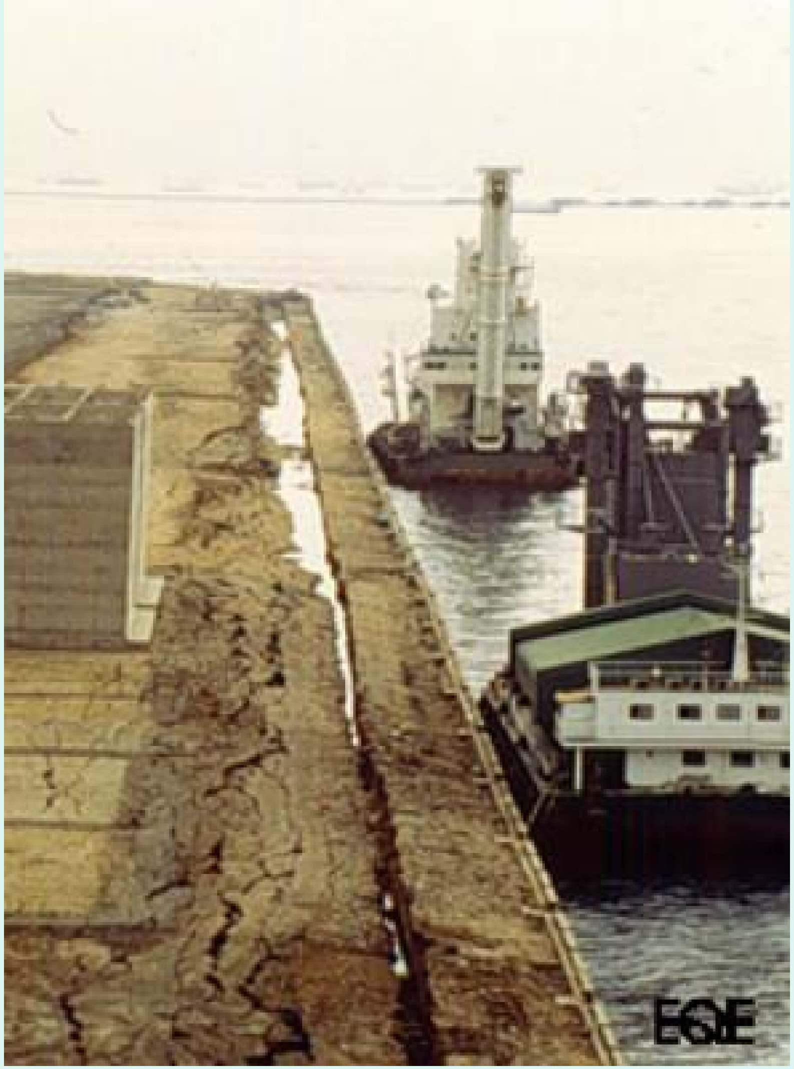
Why to learn soil mechanics ?

Engineering problems about soil

foundation liquefaction of Bansheng large earthquake

Shenghu dock:

caisson side wall slide into the sea because of blending foundation liquefaction



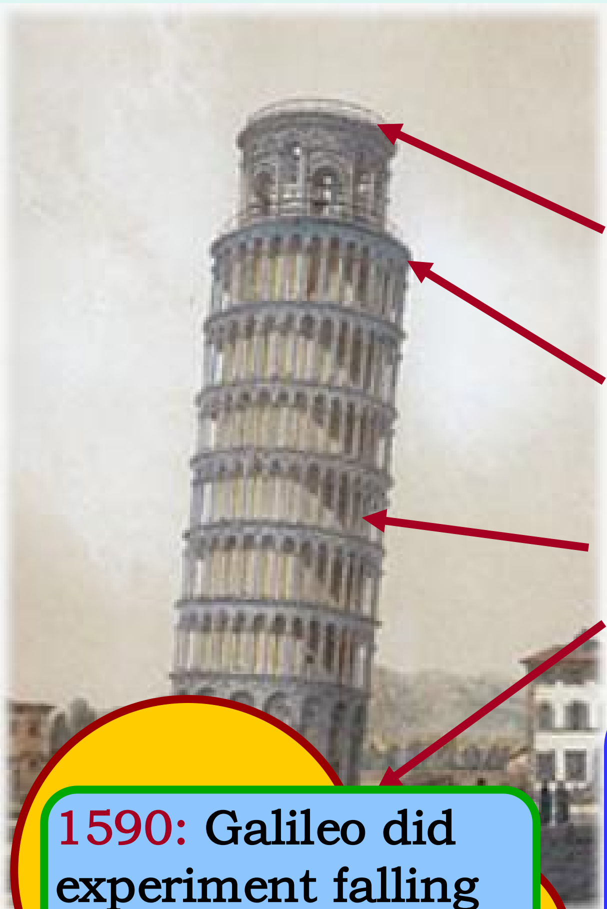
Several examples above can be attributed to **strength problem** about soil

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Pisa tip-tower



1590: Galileo did experiment falling body in the tower

now: tower leans to the south, differential settlement of south-north is 1.80m, the distance between tower top and axis has been 5.27m, the slant is 5.5°

1360: started again, completed in 1370, 8 floors, 55m highness

1272: started again, 6 years later, to 7th floor, 48m highness, stopped again

1178: to 4th floor, 29m highness, stopped because of lean

1173: started

reason:
The foundation supporting course is inorganic silt, and lower side are silt and clay blanket, its intensity is lower and distortion is bigger.

Engineering problems about soil

Pisa tip-tower

transacting method

1838-1839: loading-off through excavating ring form foundation ditch
1933-1935: foundation ditch waterproof transaction
foundation ring stabilizing grout
1990.1: sealed
1992.7: reinforced tower body, treated foundation through kentledge method and borrow method
present: has open to tourists.

Introduction

Why to learn soil mechanics？

Engineering problems about soil

Huqiu tower

survey： is on Huqiu park mountaintop in Suzhou， completed in A.D.961.
7 floors， 47.5m high， plane is octagon.

problem： tower body tilted severely, tower top is away of its axis 2.31m, there were many cracks tower bottom on its base coat， which had been dangerous building and thus closed down.

reason： is on non-uniform silty clay layer， produced non-uniform settlement.

transacting ： built a circuit of pile-row continuous underground wall around the tower and drilled inject and drive tree-root pile around tower and the tower footing in order to reinforce the tower body ， and succeeded.

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Guanxi airfield

biggest manpower island in the world

1986: start working

1990: manpower island complete

1994: airfield operation

area: 4370m×1250m

reclamation amount: 180×106m3

average thickness: 33m

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Guanxi airfield

problem：
settlement is great and non-uniform

Design estimation settlement:
5.7—7.5 m
Complete effective settlement:
8.1 m，5cm/mo.
(1990)
Time estimating complete primary consolidation ：
20 years later
More than designing extra-fill:
3m

Engineering problems about soil

several examples above can be attributed to

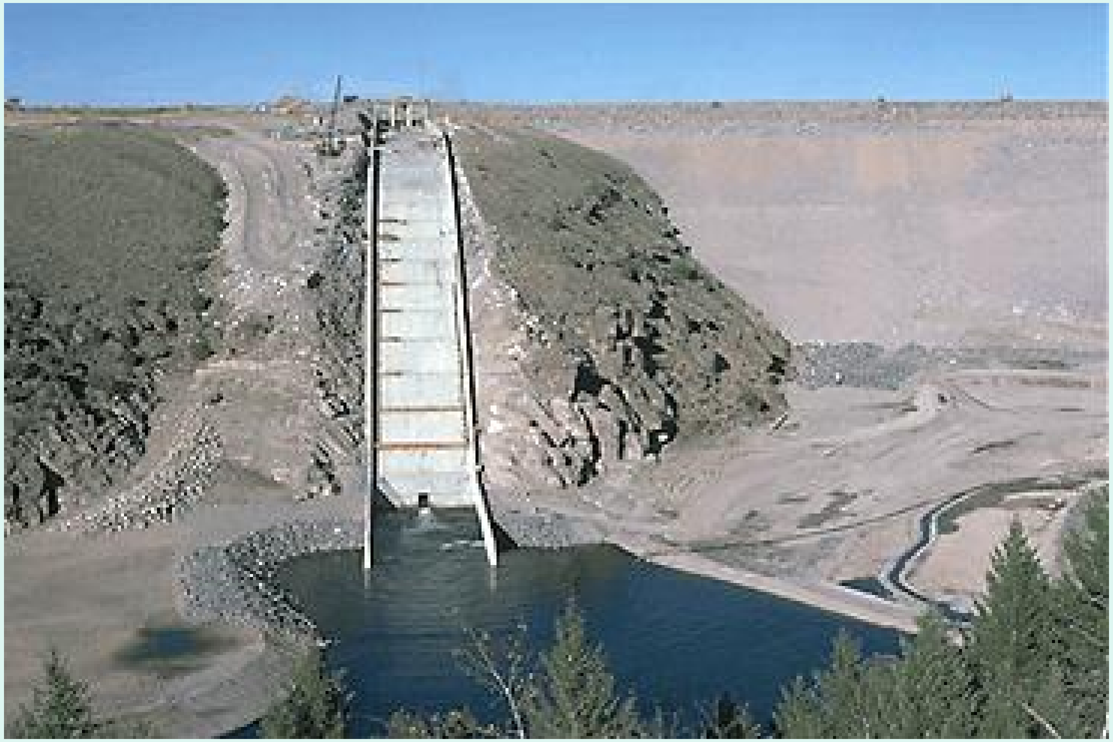
deformation problem about soil

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam



loss: lose 80 million dollar directly, 5500 indictments, 250 million dollar, 14 persons died, 25 thousand persons, 600 thousand mu land and 32 kilometres railway were stricken.

survey: earth dam, highness 90m, length 1000m, built in 1972-75, happen accident in June, 1976

reason: permeation damage—hydraulic fracture

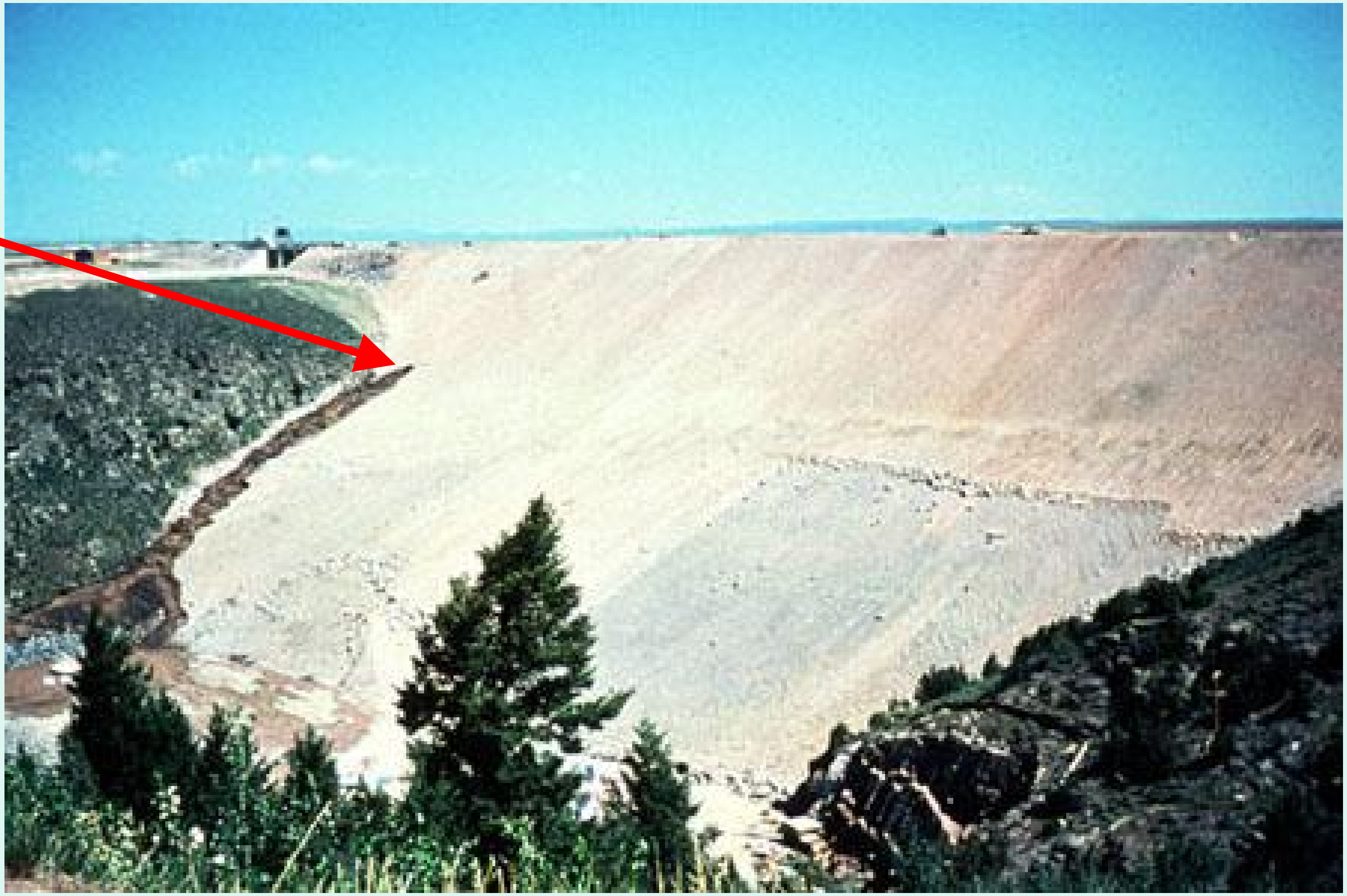
Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

At ten thirty a.m. on the fifth of June in 1976 , lower dam face has water seepage with soil.



Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

at about 11:00, cave mouth enlarged continuously and was close to dam top, and flux of mud water increased

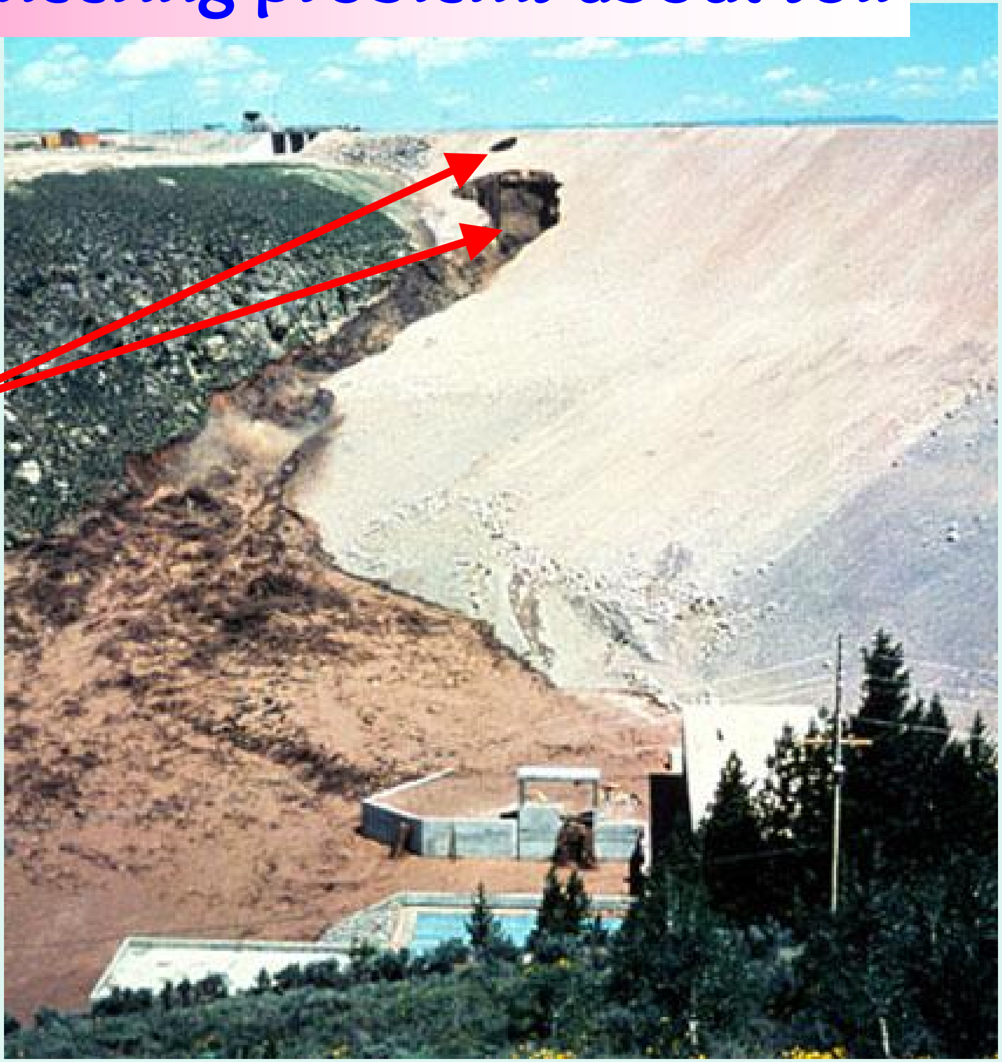
Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

At 11:30, cave mouth enlarged upwards continuously, mud water washed dam foundation, main cave top appeared a permeable-cave again. effluent mud water began to impact establishment of dam toe.



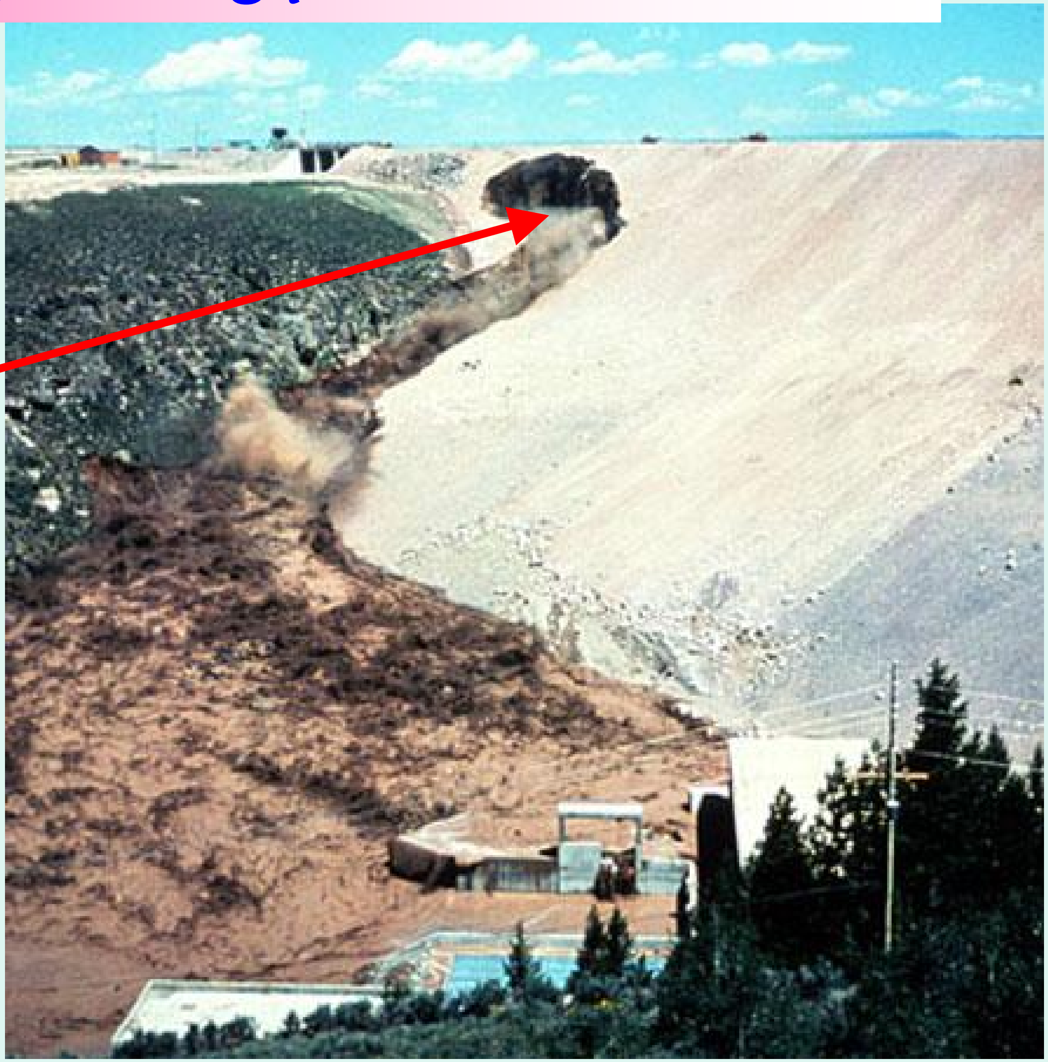
Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

At about 11: 50 cave mouth enlarged post, mud water washed dam foundation more tempestuously.



Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

at 11:57, dam slope collapsed, mud water outpoured insanely

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

collapse mouth broadened after 12:00

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

flood sweeped lowed valley bottom, all adjacent establishments were destructed thoroughly

Introduction Why to learn soil mechanics ?

Engineering problems about soil

Teton dam

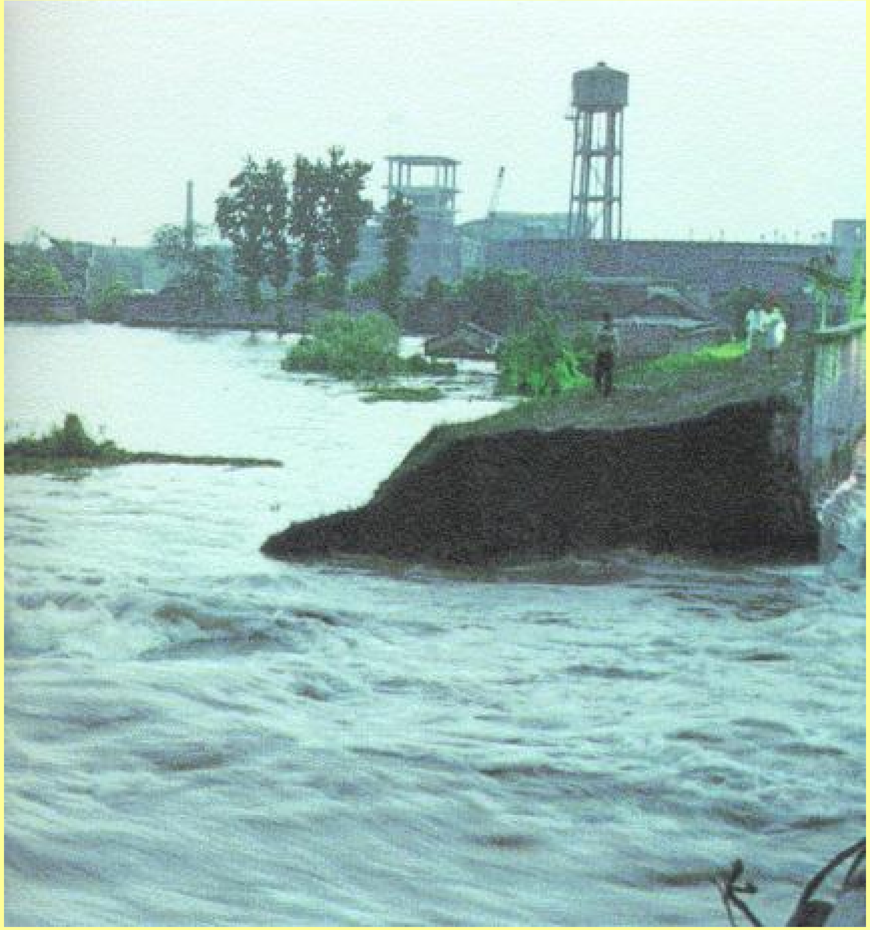
status of accident field at present

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

nine-River levee crevasse



after another 20 minutes, earth bank behind waterproof wall collapsed a small cave suddenly, with that 5 m width levee crest collapsed completely, and formed a about 62m width crevasse soon.

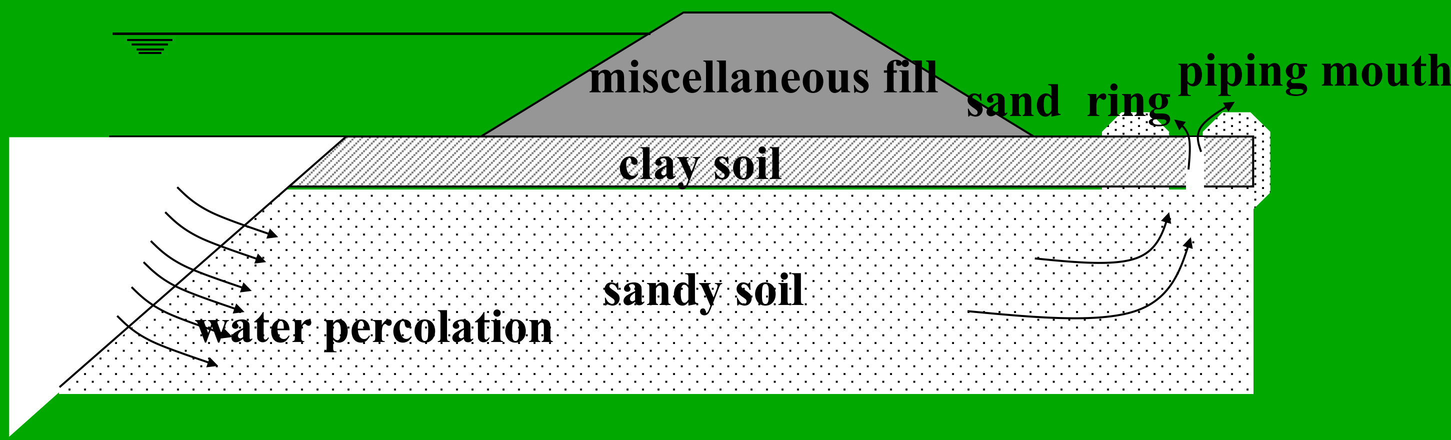
crevasse reason: levee foundation piping

Introduction

Why to learn soil mechanics ?

Engineering problems about soil

levee foundation piping



levee foundation piping happen and develop sketch map of the Changjiang River dyke building

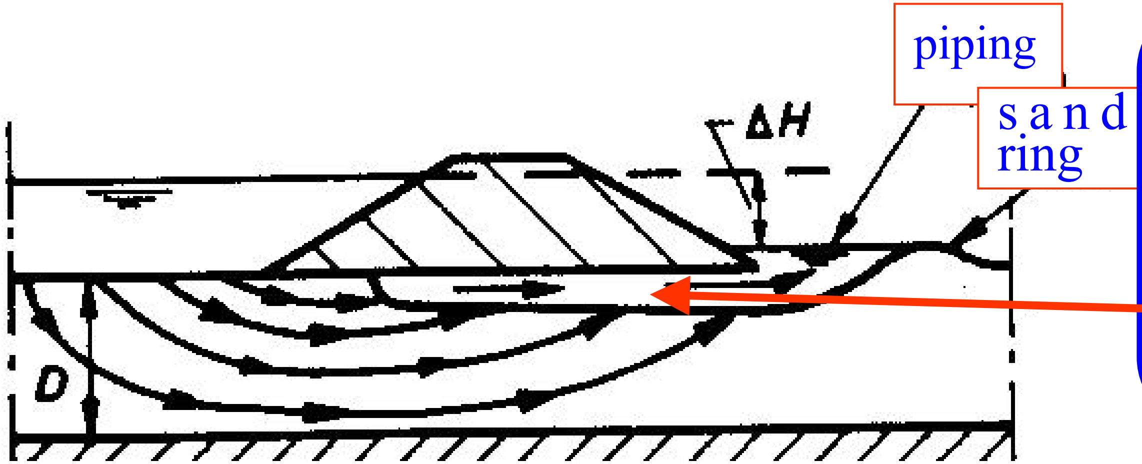
piping: on seepage action, phenomenon of fine particles in frictional soil moving or bringed out by current through pore of soil.

Introduction

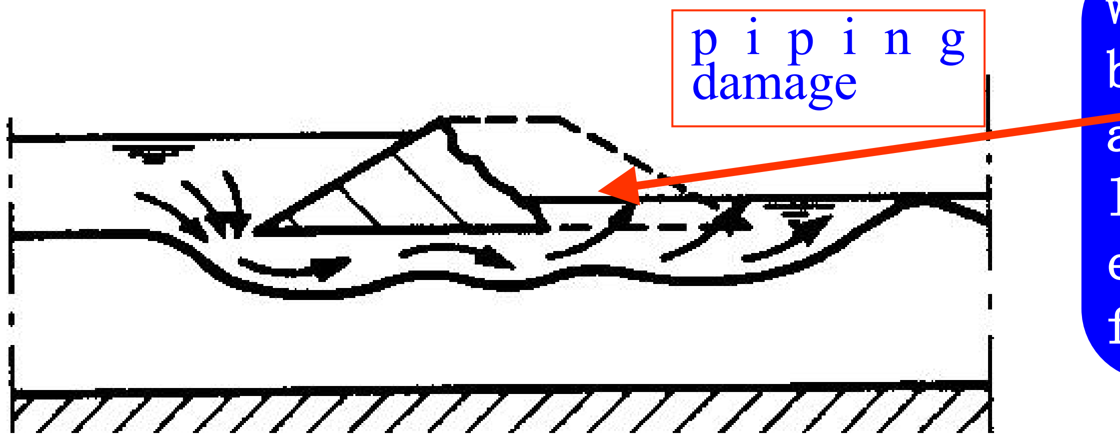
Why to learn soil mechanics ?

Engineering problems about soil

levee foundation piping damage sketch map of sandy soil



when happening piping, seepage will induce backward erosion, makes levee foundation bottom appear seepage channel



when head difference is big enough, erosion will accelerate and hollow levee foundation, and is easy to break after forming channel

Engineering problems about soil

several examples above can be attributed to **permeation problem** about soil

Introduction Why to learn soil mechanics ?

earth structure or ground

soil

strength problem
deformation problem
permeation problem

strength characteristic
deformation characteristic
permeation characteristic

Soil mechanics can solve engineering practical problems, which is existence value of soil mechanics and aim of learning soil mechanics.

Introduction

✵ What’s included by soil mechanics？

foundation

physical characteristics

forerunner

stress in soil

core

permeation characteristic

deformation characteristic

strength characteristic

specific application

soil pressure

……

Introduction What's included by soil mechanics ?

base and main line

forerunner

permeation characteristic
deformation characteristic
strength characteristic

specific appliance

1 Physical properties and classification of soil

2 Soil permeability and seepage problems

3 Calculation of stress in soil

4 Compressibility of soil and calculation of ground settlement

5 Shear strength of soil

Foundation engineering

Introduction

✵ How to learn soil mechanics well?

notice basic characteristic of soil –
through contrast with other materials

pay attention to theory contacting practice –
through field observation and test

emphasize right study method –
concept, principle, method
relation among contents
need memory , but needn’t rote

本课程安排和要求 Course schedule and requirements

教学环节　teaching link

课堂讲授（22 次 44 学时）

classroom teaching

实验课　（6 次　12 学时）

laboratory course

课堂表现及作业

考核及成绩

examination and grade

70%　（期末考试）　**course grade**

　final examination 80%

　experiment grade 20%

30%　（作业＋表现＋ 出勤）

assignment+performance+attendance

要求 Requirements

1. 大课必须出席，不定期点名。 Attendance required.
2. 上课认真听讲，积极回答问题。 listen carefully, actively answer the question.
3. 试验：注意观察，联系理论。Do the experiment carefully.
4. 作业：独立完成，每章交一次。Hand in homework on time.

See you next time,
welcome to ask we questions!